AWS Application Load Balancer (ALB) Practice Tasks

Task 1: Basic Load Balancer

Objective: Learn this ALB feature.

Description: Launch two EC2 instances, install Apache/Nginx, configure different home pages, create a Target Group, register both instances, create an ALB, attach the Target Group, and verify requests are distributed between both servers by refreshing the ALB DNS.

Task 2: Health Checks

Objective: Learn this ALB feature.

Description: Stop the web server on one instance and observe it become unhealthy. Verify traffic only reaches the healthy instance. Restart the service and confirm it becomes healthy again.

Task 3: Path-Based Routing

Objective: Learn this ALB feature.

Description: Create two target groups and configure listener rules so /app1 goes to one target group and /app2 goes to another.

Task 4: Host-Based Routing

Objective: Learn this ALB feature.

Description: Use two hostnames and configure listener rules so each hostname routes to a different target group.

Task 5: Add New Server

Objective: Learn this ALB feature.

Description: Launch a third EC2 instance, install the application, add it to the target group and verify traffic reaches all three instances.

Task 6: Deregister Instance

Objective: Learn this ALB feature.

Description: Remove one instance from the target group and verify traffic is only served by the remaining instances.

Task 7: Sticky Sessions

Objective: Learn this ALB feature.

Description: Enable stickiness on the target group and verify repeated requests from the same browser reach the same backend.

Task 8: HTTPS

Objective: Learn this ALB feature.

Description: Create an ACM certificate, configure HTTPS listener on the ALB and redirect HTTP to HTTPS.

Task 9: Multi-Port Routing

Objective: Learn this ALB feature.

Description: Run applications on different ports and configure target groups and listener rules accordingly.

Task 10: Access Logs

Objective: Learn this ALB feature.

Description: Enable ALB access logs to an S3 bucket, generate traffic and inspect the log files.

Task 11: Security Groups

Objective: Learn this ALB feature.

Description: Allow internet traffic only to the ALB and allow EC2 instances to accept traffic only from the ALB security group.

Task 12: Auto Scaling

Objective: Learn this ALB feature.

Description: Create a Launch Template and Auto Scaling Group attached to the ALB target group. Increase desired capacity and verify new instances are automatically registered.